INTRODUCTION TO APPLIED NUMERICAL METHODS

Numerical Integration: Trapezoidal Rule

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BACKGROUND

Numerical Integration

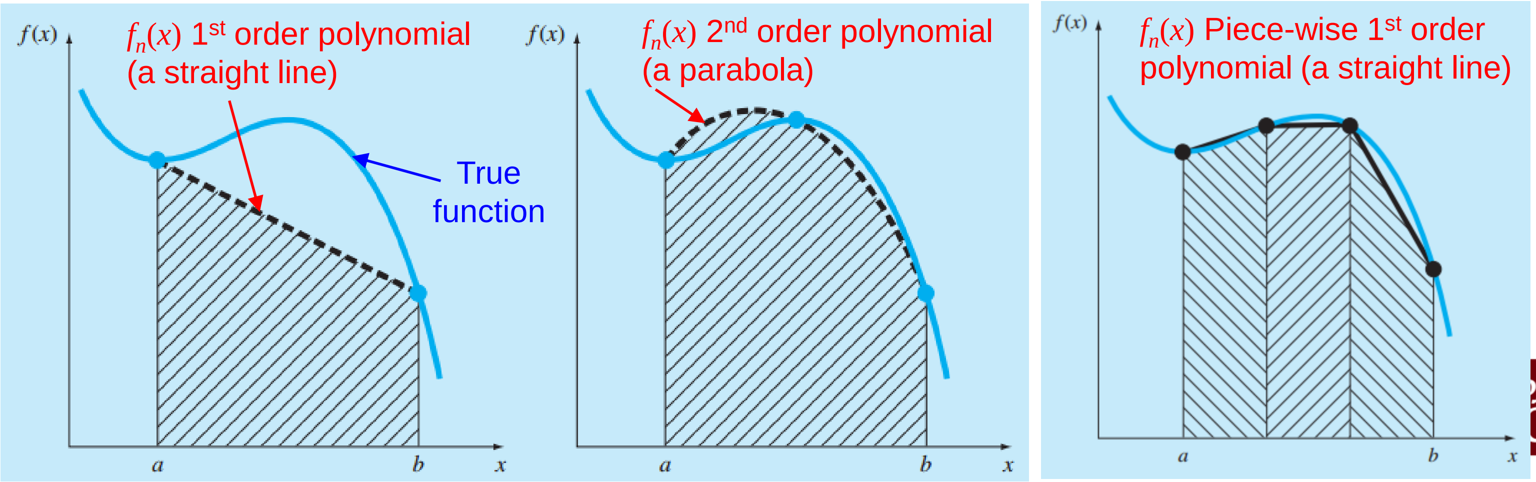
- Replacing a true (but unknown) or a complicated function or tabulated data with an approximating function

$$I = \int_a^b f(x)\,dx \cong \int_a^b f_n(x)\,dx$$

True (but unknown) or complicated function

An approximating function

$f_n(x)$ is in a polynomial form: $\boxed{f_n(x) = a_0 + a_1 x + \cdots + a_{n-1}x^{n-1} + a_n x^n}$ n: order of the polynomial

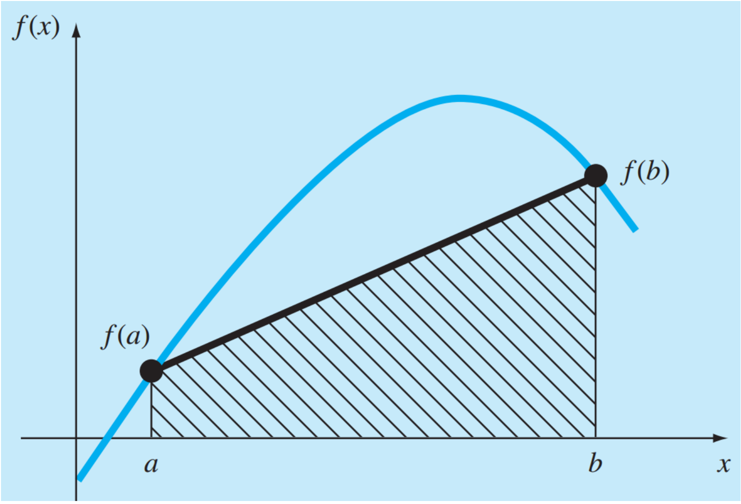


TRAPEZOIDAL RULE

The Trapezoidal Rule:

- The easiest scheme for numerical integration
- The approximating polynomial is the first-order (a line)

$$\boxed{I = \int_a^b f(x)\,dx \cong \int_a^b f_1(x)\,dx}$$



- For a linear interpolation within x = [a b]

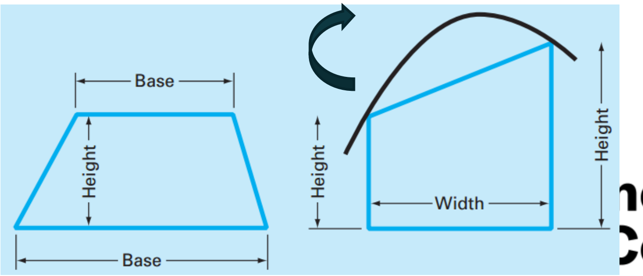
$$\boxed{f_1(x) = f(a) + \frac{f(b) - f(a)}{b - a}(x - a)}$$

Substitute $f_1(x)$ into the integral I

$$\boxed{I = \int_a^b \left[f(a) + \frac{f(b) - f(a)}{b - a}(x - a) \right] dx}$$

Trick: define $\tilde{x} = x - a$ and $\tilde{x}$ in the range $[0, b-a]$

$$I = (b-a)\frac{f(a) + f(b)}{2}$$



Trapezoidal rule → Equivalent to approximating area of the trapezoid

DERIVATION OF TRAPEZOIDAL RULE

$$I = \int_a^b f(x)\,dx \cong \int_a^b f_1(x)\,dx$$

Expand $f_1(x)$:

$$f_1(x) = \frac{f(b) - f(a)}{b - a} x + f(a) - \frac{af(b) - af(a)}{b - a}$$

Grouping the last two terms:

$$f_1(x) = \frac{f(b) - f(a)}{b - a} x + \frac{bf(a) - af(b)}{b - a}$$

Integrated between $x = a$ & $x = b$:

$$I = \frac{f(b) - f(a)}{b - a} \frac{x^2}{2} + \frac{bf(a) - af(b)}{b - a} x \Big|_b^a$$

$$f_1(x) = f(a) + \frac{f(b) - f(a)}{b - a}(x - a)$$

Reform the integration:

$$I = \frac{f(b) - f(a)}{b - a} \frac{(b^2 - a^2)}{2} + \frac{bf(a) - af(b)}{b - a}(b - a)$$

Given $b^2 - a^2 = (b - a)(b + a)$:

$$I = [f(b) - f(a)]\frac{b + a}{2} + bf(a) - af(b)$$

Multiplying and collecting herms:

$$I = (b - a)\frac{f(a) + f(b)}{2}$$

AN EXAMPLE

Using the Trapezoidal rule to numerically integrate

$$f(x) = 0.2 + 25x - 200x^2 + 675x^3 - 900x^4 + 400x^5$$

From *a* = 0 and *b* = 0.8 (the exact value of integral is 1.640533).

Solution:

$$I = (b-a)\frac{f(a)+f(b)}{2}$$

$$f(0) = 0.2$$

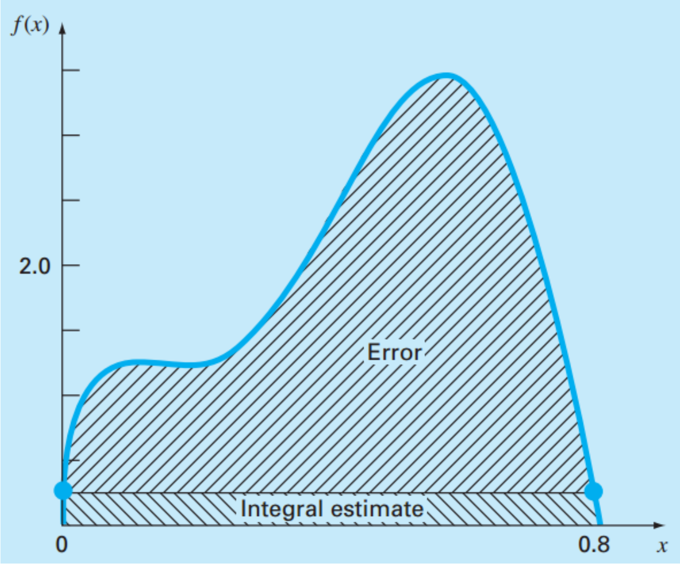
$$f(0.8) = 0.232$$

$$a = 0$$

$$b = 0.8$$

$$\boxed{I = 0.8\frac{0.2+0.232}{2} = 0.1728}$$

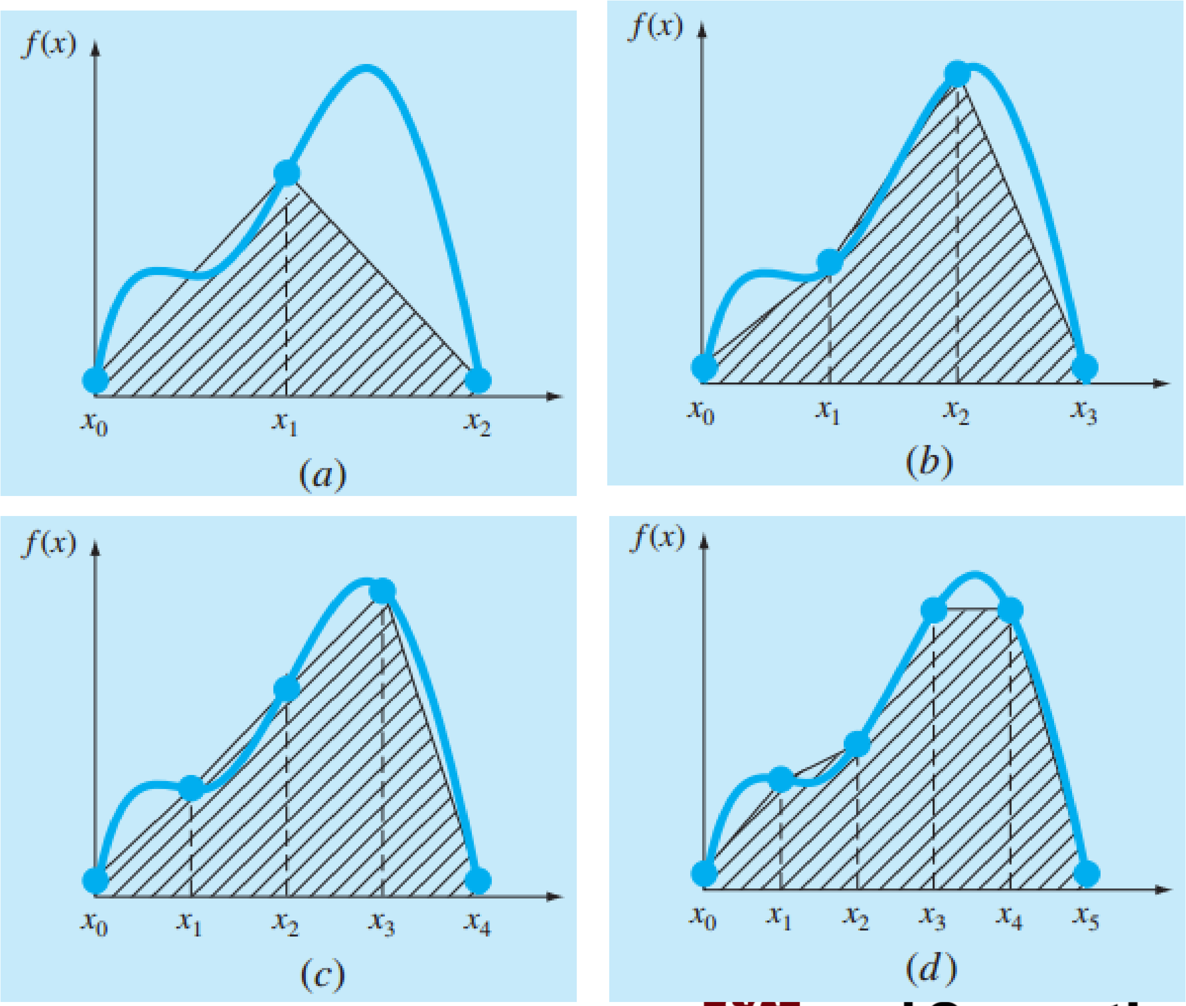
$$\boxed{\begin{aligned} E_t &= 1.640533 - 0.1728 = 1.467733 \\ \varepsilon_t &= 89.5\% \end{aligned}}$$



The data points are too far apart causing mis-representation of the function (and the portion of the integral above the line contributes large error)

THE MULTIPLE-APPLICATION TRAPEZOIDAL RULE

- **Improving the Single Application of the Trapezoidal Rule:**
- Divide the integration interval into a number of segments/sub-intervals
- Apply the trapezoidal rule to each segment



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THE MULTIPLE-APPLICATION TRAPEZOIDAL RULE

- Divide the integration interval into *n* segments (n+1 base points)
- Each segment

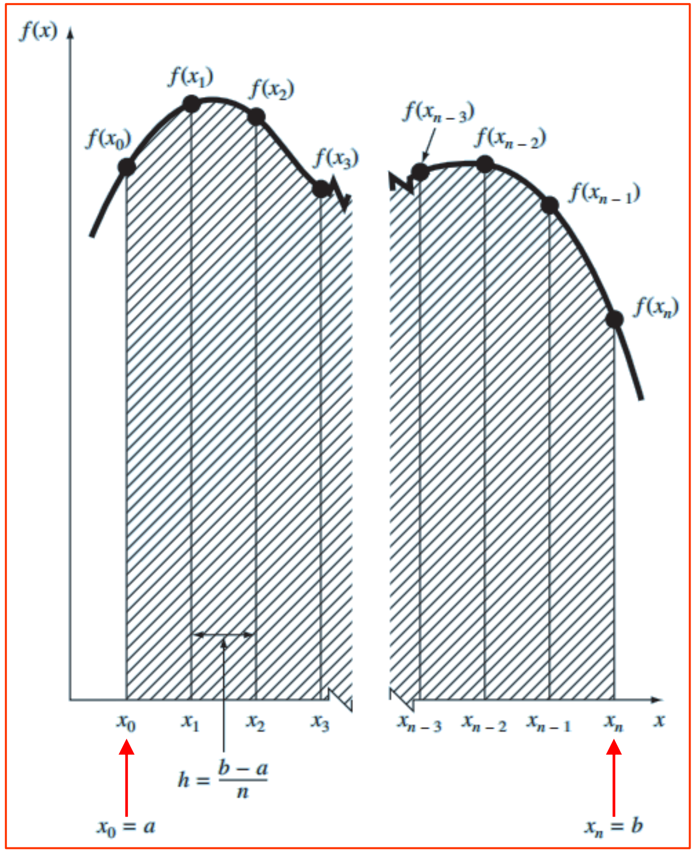
$$\boxed{h = \frac{b-a}{n}}$$

$$\boxed{I = \int_{x_0}^{x_1} f(x)\,dx + \int_{x_1}^{x_2} f(x)\,dx + \cdots + \int_{x_{n-1}}^{x_n} f(x)\,dx}$$

$$\boxed{I = h\frac{f(x_0)+f(x_1)}{2} + h\frac{f(x_1)+f(x_2)}{2} + \cdots + h\frac{f(x_{n-1})+f(x_n)}{2}}$$

$$\boxed{I = \frac{h}{2}\left[f(x_0) + 2\sum_{i=1}^{n-1} f(x_i) + f(x_n)\right]}$$

$$I = \underbrace{(b-a)}_{\text{Width}}\underbrace{\frac{f(x_0) + 2\sum_{i=1}^{n-1} f(x_i) + f(x_n)}{2n}}_{\text{Average height}}$$



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AN EXAMPLE

Using the two-segment Trapezoidal rule to numerically integrate

$$f(x) = 0.2 + 25x - 200x^2 + 675x^3 - 900x^4 + 400x^5$$

From a = 0 and b = 0.8 (the exact value of integral is 1.640533).

Solution:

$n = 2$ (3 data points), $h = 0.4$

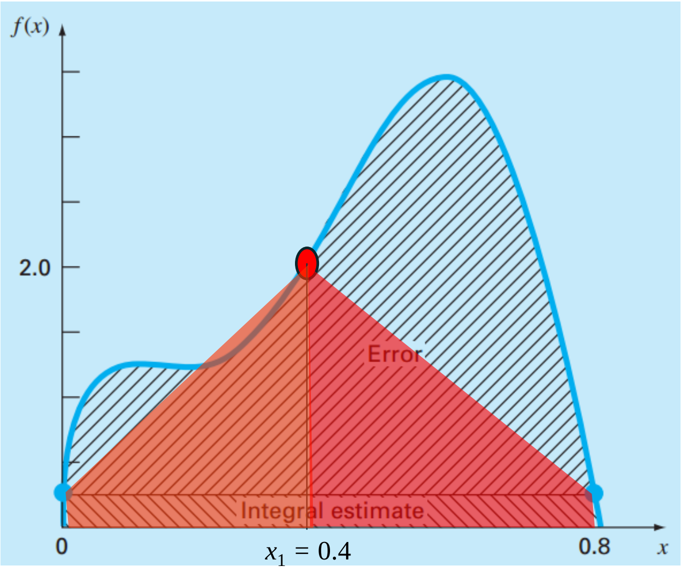
$x_0 = a = 0 \qquad f(0) = 0.2$

$x_2 = b = 0.8 \qquad f(0.8) = 0.232$

$x_1 = 0.4 \qquad f(0.4) = 2.456$

$$I = \underbrace{(b-a)}_{\text{Width}} \underbrace{\frac{f(x_0) + 2\sum_{i=1}^{n-1} f(x_i) + f(x_n)}{2n}}_{\text{Average height}}$$

$$\boxed{I = 0.8\frac{0.2 + 2(2.456) + 0.232}{4} = 1.0688}$$



$$\boxed{E_t = 1.640533 - 1.0688 = 0.57173 \qquad \varepsilon_t = 34.9\%}$$

Using multi-segment trapezoidal rule significantly improves the integration accuracy